Survey Design Analysis of NHANES 2015–2016

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Dataset Background

NHANES Dataset:

- 2015-2016 wave of the National Health and Nutrition Examination Survey (NHANES)
- Cross-sectional survey of randomly selected U.S. communities
- Records thousands of phenotypes (age, sex, BMI, systolic blood pressure, etc.)

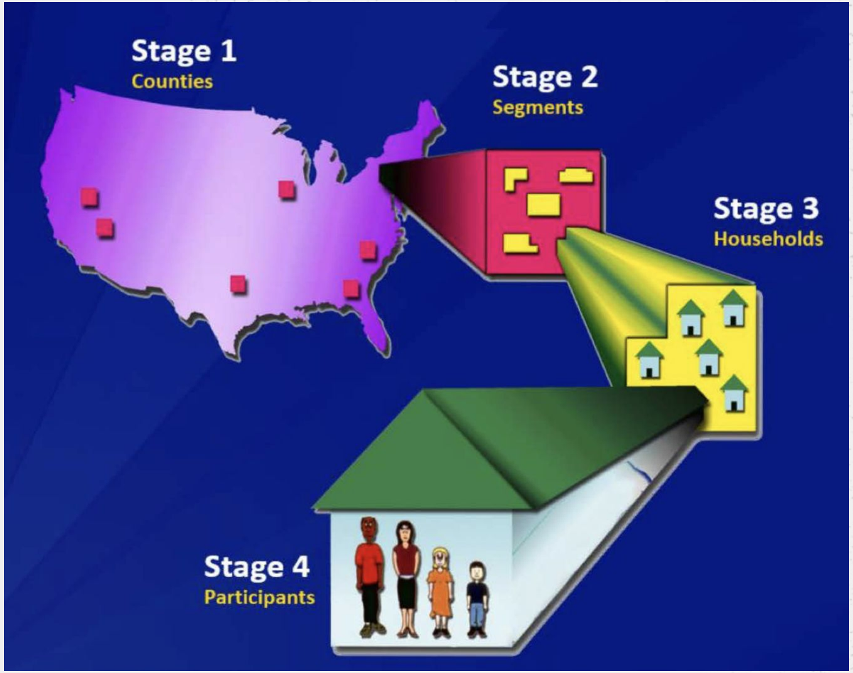
Goals of NHANES:

- Monitor trends in selected diseases, risk behaviors and environmental exposures in the U.S. population
- Study the relationship between diet, nutrition, and health;
- Explore emerging public health issues and new technologies; and
- Provide baseline health characteristics that can be linked to mortality data

nhanes National Health and Nutrition Examination Survey

Project Aim

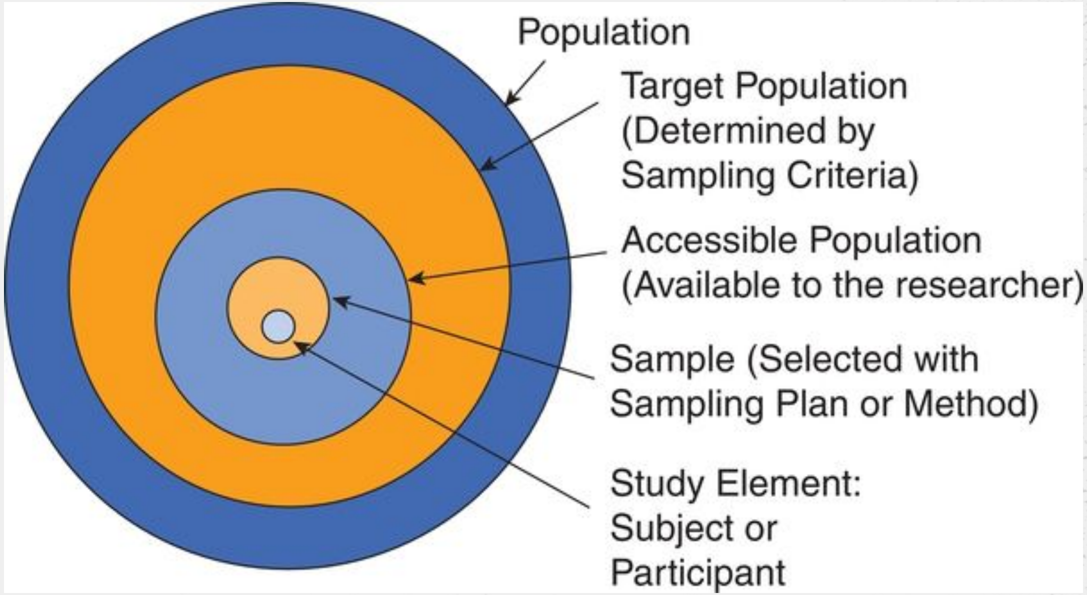
- Analyze a real-world complex survey
- Produce finite population inferences
- Evaluate sampling design affects on precision and efficiency



Population and Elements

Target population:

noninstitutionalized U.S. population

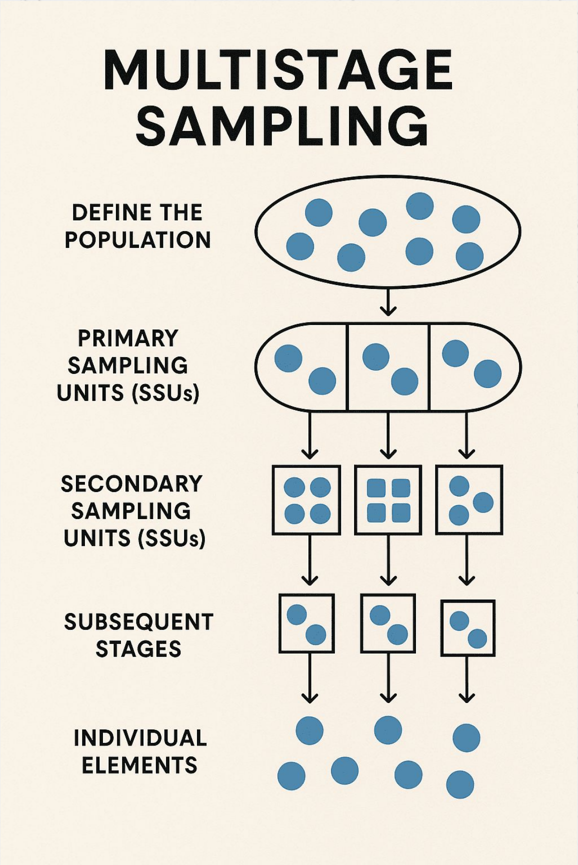


Elements: individuals in the NHANES 2015-2016 sample

Population Size: not directly observed

Sampling Frame and Design

- **Multistage Probability Sampling Design**
  - Counties as Primary Sampling Units (PSUs)
  - Randomly select Secondary Sampling Units (SSUs)
  - Sample random units within the SSUs
- **Features**
  - Stratification
  - Clustering
  - Oversampling (minorities, elderly)



Sample and Weights

	unweighted_n <int>	population_estimate <dbl>	SE <dbl>	lower_95 <dbl>	upper_95 <dbl>
one	7256	271045833	15184674	241284418	300807248

Weighting the sample allows the researchers to adjust for:

- unequal selection probability
- Nonresponse
- population calibration.

Estimation Methods

Using R programming language (survey package):

- Mean: svymean()
- Proportion: svymean(hi_tchol)
- Ratio: svyratio()
- Regression: svyglm()

Each of these methods incorporate weights, and account for stratification and clustering.

Key Results

Mean Cholesterol
svymean(~lbxtc)

	mean	SE	2.5%	97.5%
lbxtc	185.9281	1.265995	183.4468	188.4094

Proportion with High Cholesterol
svymean(~hi_tchol)

	mean	SE	2.5%	97.5%
hi_tchol	0.10429	0.0074	0.08974905	0.1188357

Ratio Estimator
svyratio(~lbxtc, ~ridageyr)

	mean	SE	2.5%	97.5%
lbxtc/ridageyr	4.447244	0.05803922	4.33349	4.560999

Design Effect and Efficiency

naive_mean	naive_se	design_mean.lbxtc	design_se
180.2566152	0.4852786	185.928123	1.265995

- svymean(..., deff = TRUE) compares the variance under the NHANES design to the variance under a simple random sample
- **DEFF > 1** indicates that the complex survey design increases variance

Main sources of increased variance:

- **Clustering within PSUs**
- **Unequal survey weights**

Interpretation: NHANES is less statistically efficient than an SRS, but it produces valid national inferences

- Ignoring the design would understate uncertainty and overstate precision

Nonresponse and Bias

Types of Nonresponse

- Unit nonresponse
  - No responses from selected unit
- Item nonresponse
  - Unit only answers some but not all survey questions

NHANES Adjustments

- Weighting adjustments

Estimate Type	Mean Cholesterol	SE
Naive (ignores weights)	180.26	0.49
NHANES weighted	185.93	1.27

Note: Potential for Bias if nonresponse is not random.

Conclusion

NHANES Strengths

- Nationally representative
- Rich data

NHANES Limitations

- Complex design can frequently cause higher variance
- Nonresponse and coverage issues

NHANES provides high-quality, representative data, but requires proper design-based analysis. Ignoring survey design can lead to incorrect inference.

Resources

Damico, A. (n.d.). *NHANES Survey Data Analysis Example*. Retrieved from https://asdfree.com/national-health-and-nutrition-examination-survey-nhanes.html